Hi-Engineer

Trade-show & media pipeline investigation

Full audit of the input source list, curl-vs-Firecrawl fetch strategy, downstream jobs and news pipelines, and the shape of the data collected to date.

INPUT LISTS	COMPANIES	JOBS EXTRACTED	NEWS EXTRACTED
35	**1,190**	**930**	**2,328**
26 trade-show + 9 media	in the directory	across all companies	across all companies

ENRICHED	WITH SITE	JOBS URL	NEWS URL
889	**783**	**430**	**516**
75% of total	66% coverage	55% of sited	66% of sited

Generated 2026-08-24 · Data source: production Heroku Postgres (`hi-engineer-app`)

1 · Executive summary

The Hi-Engineer platform ingests two input lists — **26 trade-show exhibitor pages** and **9 industrial/food media portals** — normalises everything into a single company directory, cascades through four enrichment providers, then runs **jobs** and **news** pipelines against each company's own site.

The trade-show ingest is **fully wired and productive**. Of the 26 event sources, 14 are fetched with a plain `requests` call (foodtech + easyfairs parsers), 9 require Firecrawl's JS-render tier, and 3 are known-blocked and kept in the table only so the reason is visible in the UI. The current directory holds **1,190 companies** (889 enriched, 783 with a discovered website).

The media portal ingest is **largely blocked**. All 9 portals were probed with direct HTTP; every one of them returned zero usable entries — five build their directory in the browser (JavaScript/AJAX), one blocks scrapers at the edge (HTTP 403), and three have parser mis-matches that need custom rules. Firecrawl would unlock the JS-only tier at a marginal cost (details in §4).

Downstream, the platform has captured **930 job listings** and **2,328 news articles** across 1,190 companies. Coverage is heavily skewed towards companies that publish structured careers/blog pages: the top 5 companies account for 213 jobs (23%) and 436 articles (19%) of the totals.

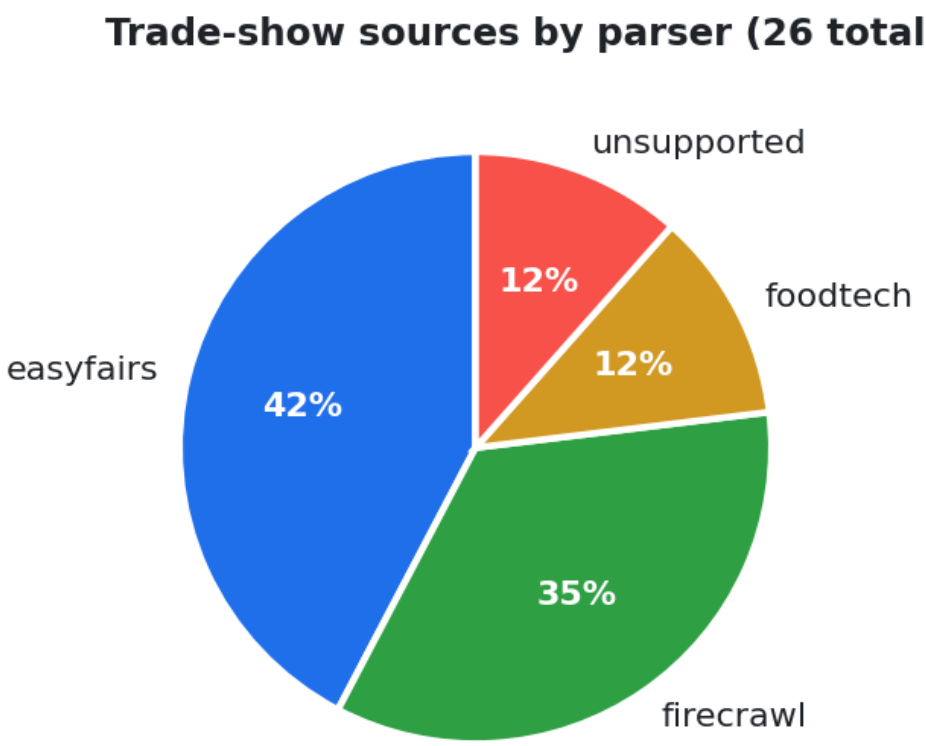
Key numbers at a glance

TRADE-SHOW INPUTS	MEDIA INPUTS	DIRECT-CURL USABLE	FIRECRAWL REQUIRED
26 6 parsers	**9** 1 parser family	**14** 54% of trade shows	**14** 9 shows + 5 media

FIRECRAWL BALANCE	COMPANIES WITH A WEBSITE	JOBS PAGE DISCOVERED	NEWS PAGE DISCOVERED
1,525 of 1,000/mo	**783** 66%	**430** 55% of sited	**516** 66% of sited

2 · Source inventory

Trade shows: 26 URLs across 4 parser families. Each row in the `event_sources` table records a parser hint, the last scrape count, and the last error. The chart below shows the parser split; the table on the next page is the full inventory.



Parser families explained

foodtech — pages built on the `data-component="card-introduce"` layout (FoodTech Event, MT Conference, Automation Xperience). One direct HTTP call recovers the entire list.
easyfairs — SSR anchor pattern (Empack, Rotterdam Processing Week, Aqua Nederland, ReMaTec, Interclean, etc.). Only the first ~25 exhibitors are in the static HTML; the rest is behind an authenticated backend, so we capture the visible slice and derive clean names from URL slugs.
firecrawl — JS-rendered exhibitor lists (Provada, TechniShow, BouwBeurs, VSK, ESEF, InfraTech, Maintenance NEXT, WoTS, Foodspec). Rendered in a headless browser via the Firecrawl API, then run through one of five DOM extractors (*provada*, *stand_table*, *foodtech*, *safetyevent*, *generic_anchor*).
unsupported — LinkedIn, Vakbeurs Energie, Safety Event. Kept in the table so the UI shows the reason; no fetch attempt.

Media portals: 9 URLs, 1 parser dispatcher. Each portal has a per-domain rule (four written), with a noise-filtered fallback for the rest. Every portal was tried with direct HTTP and all currently return zero entries — see §3.2 for the reason breakdown.

2.1 · Trade-show source inventory (full)

Sorted by parser then by last exhibitor count. *Last count* is the number of companies extracted on the most recent scrape.

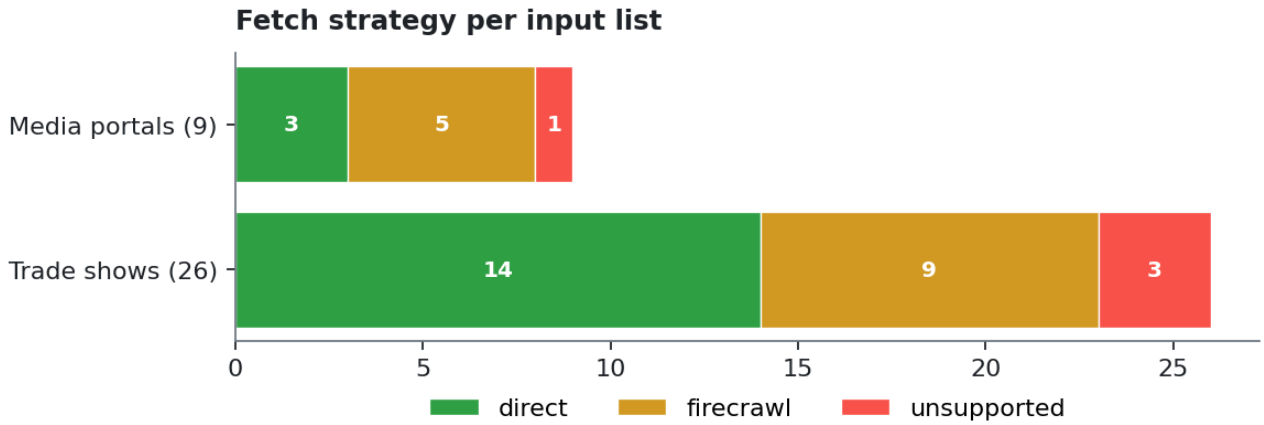
Label	Parser	Last count	Last scraped	Last error
ReMaTec Amsterdam (Apr 2027) — remanufa…	easyfairs	20	2026-07-16	—
Interclean Amsterdam — cleaning industry	easyfairs	20	2026-07-16	—
Rotterdam Processing Week (Easyfairs)	easyfairs	16	2026-07-16	—
Solids Rotterdam 2027 (Easyfairs)	easyfairs	15	2026-07-16	—
Empack (Easyfairs)	easyfairs	15	2026-07-16	—
Bakkersvak & IJs-Vak (Easyfairs)	easyfairs	15	2026-07-16	—
De Groene Sector Vakbeurs (Easyfairs)	easyfairs	15	2026-07-16	—
Dutch Pork & Poultry Expo (Easyfairs)	easyfairs	15	2026-07-16	—
Pumps & Valves Rotterdam 2027 (Easyfair…	easyfairs	15	2026-07-16	—
M+R Rotterdam 2027 (Easyfairs)	easyfairs	7	2026-07-16	—
Aqua Nederland (Easyfairs)	easyfairs	0	2026-07-16	—
TechniShow 2026 (Jaarbeurs) — industria…	firecrawl	327	2026-07-16	—
BouwBeurs 2027 (Jaarbeurs) — constructi…	firecrawl	275	2026-07-16	—
VSK+E 2026 (Jaarbeurs) — installation i…	firecrawl	159	2026-07-16	—
Provada	firecrawl	12	2026-07-16	—
InfraTech 2027 (Ahoy) — infrastructure	firecrawl	2	2026-07-16	—
Vakbeurs Foodspecialiteiten 2026 — food…	firecrawl	1	2026-07-16	—
ESEF Maakindustrie 2026 (Jaarbeurs) — m…	firecrawl	0	2026-07-16	—
Maintenance NEXT 2027 (Ahoy) — asset ma…	firecrawl	0	2026-07-16	—
WoTS 2026 (Jaarbeurs, via FHI) — World…	firecrawl	0	2026-07-16	—
Food Tech Event	foodtech	117	2026-07-16	—
Automation Xperience (Apr 2027, Gorinch…	foodtech	104	2026-07-16	—
MT Conference	foodtech	91	2026-07-16	—
Safety Event (partners)	unsupported	1	2026-07-14	partners loaded via JS…
Mikrocentrum (LinkedIn)	unsupported	—	—	—
Vakbeurs Energie	unsupported	—	—	—

Media portals — same information for the content-engine input list. Every portal has been probed; none are currently producing usable output via direct HTTP.

Portal (host)	Status	Reason
installatieenbouw.nl	no_entries	empty result; parser returned 0
vakbladvoedingsindustrie.nl	http_403	403 Forbidden (blocked)
industriebouw-online.nl	no_entries	empty result
processcontrol.nl	js_only	JS/AJAX directory — page shell only
solidsprocessing.nl	js_only	JS/AJAX directory
fluidsprocessing.nl	js_only	JS/AJAX directory
industrielinqs.nl	js_only	JS/AJAX directory
regiobedrijf.nl	no_entries	empty result
tim.pmg.nl	js_only	JS/AJAX directory

3 · Direct cURL vs Firecrawl

The pipeline defaults to a plain HTTP fetch and reaches for Firecrawl only when the DOM is empty without a JavaScript runtime. This section splits each input list by the strategy actually in use.



3.1 · Trade shows — 14 of 26 fetched with plain HTTP

Of the 26 event-source rows, **14 are direct-HTTP capable** (foodtech + easyfairs + safetyevent parsers), **9 require Firecrawl** (all JS-rendered Jaarbeurs / Ahoy / FHI / Provada portals), and **3 are architecturally blocked** (LinkedIn requires an account; Vakbeurs Energie and Safety Event ship an empty shell that no headless browser has been able to populate).

3.2 · Media portals — 0 of 9 currently productive

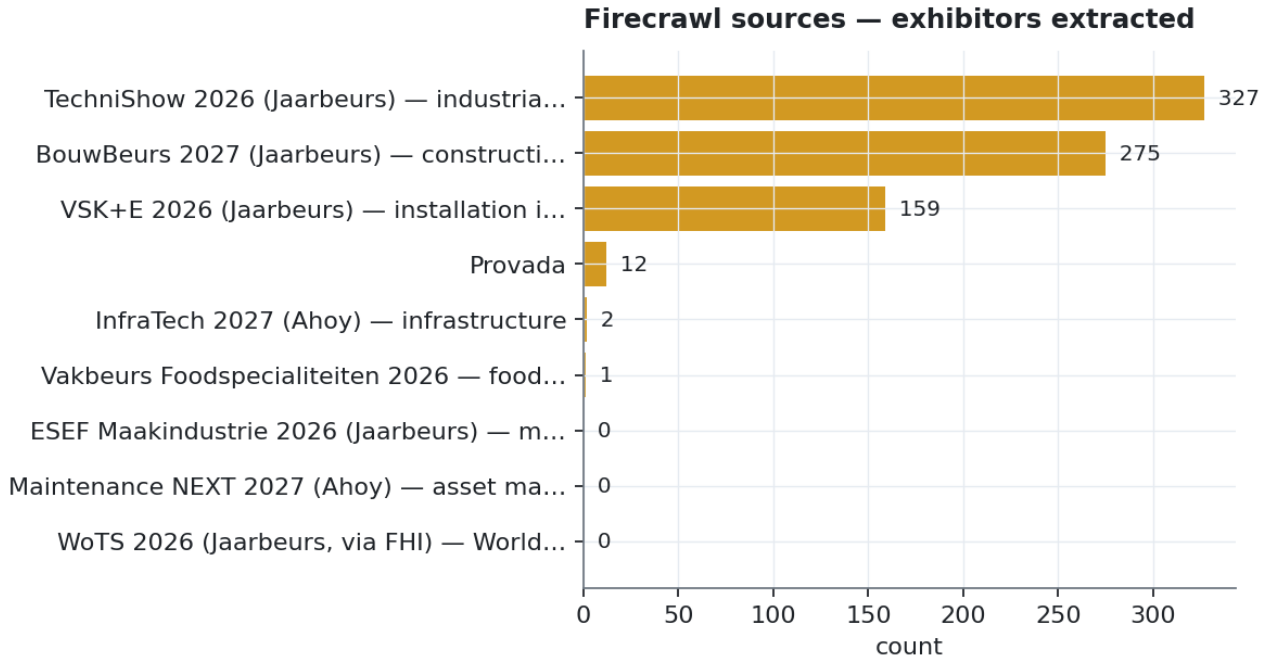
Every portal was probed with a normal browser User-Agent and returned zero entries. Grouped by cause:

Category	#	Portals
JS/AJAX only (would work through Firecrawl)	5	processcontrol.nl, solidsprocessing.nl, fluidsprocessing.nl, industrielinqs.nl, tim.pmg.nl
HTTP 403 (edge-blocked)	1	vakbladvoedingsindustrie.nl
Zero entries after per-domain parser	3	installatieenbouw.nl, industriebouw-online.nl, regiobedrijf.nl

4 · Firecrawl status and pricing

REMAINING CREDITS	PLAN CREDITS / MO	SOURCES CURRENTLY ON FIRECRAWL	WOULD-ADD IF MEDIA ENABLED
1,525 in period ending 2026-09-23	**1,000** renews monthly on the 23rd	**9** trade shows only	**5** JS-only media portals

4.1 · Yield per Firecrawl source (last run)



The *stand_table* extractor added in the last release unlocked TechniShow, BouwBeurs, and VSK — the three big Jaarbeurs sites where the exhibitor list is rendered as a single-cell <td> per exhibitor. **ESEF and Maintenance NEXT still return zero**: ESEF renders a hub page that links out to sub-pages, and Maintenance NEXT renders almost nothing at all (probably a cookie/consent wall). WoTS, InfraTech, and Foodspec pick up a handful each through the generic anchor extractor but the extractor is clearly under-tuned for their DOMs.

4.2 · Cost to run everything at planned cadences

One Firecrawl scrape = one credit. At today's mix (9 JS-only trade-show sources), a single full pass of the pipeline costs **9 credits** (≈ $0.144 at the current $16/1,000 rate).

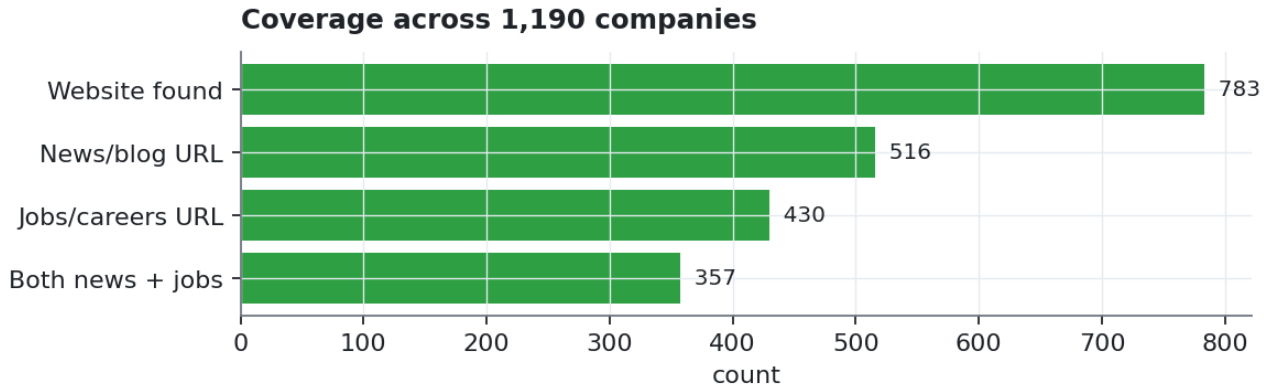
Cadence	Trade-shows only	+ Media (JS-only)	Fits current plan?
Daily	270 cr/mo	420 cr/mo	Yes
Weekly	36 cr/mo	56 cr/mo	Yes (easily)
Every 2 days	135 cr/mo	210 cr/mo	Yes
Hourly (extreme)	6480 cr/mo	10080 cr/mo	No — upgrade needed

Bottom line: at any realistic cadence (weekly to daily), the current 1,000-credit plan covers both the trade show set and the five JS-only media portals with headroom. Only an hourly run would push us into an upgrade tier.

Pricing is billed by Firecrawl per credit; the illustrative $16/1,000 rate above matches the published Hobby-tier price. The account here reports a 1,000-credit monthly plan with rollover — details in the Firecrawl dashboard.

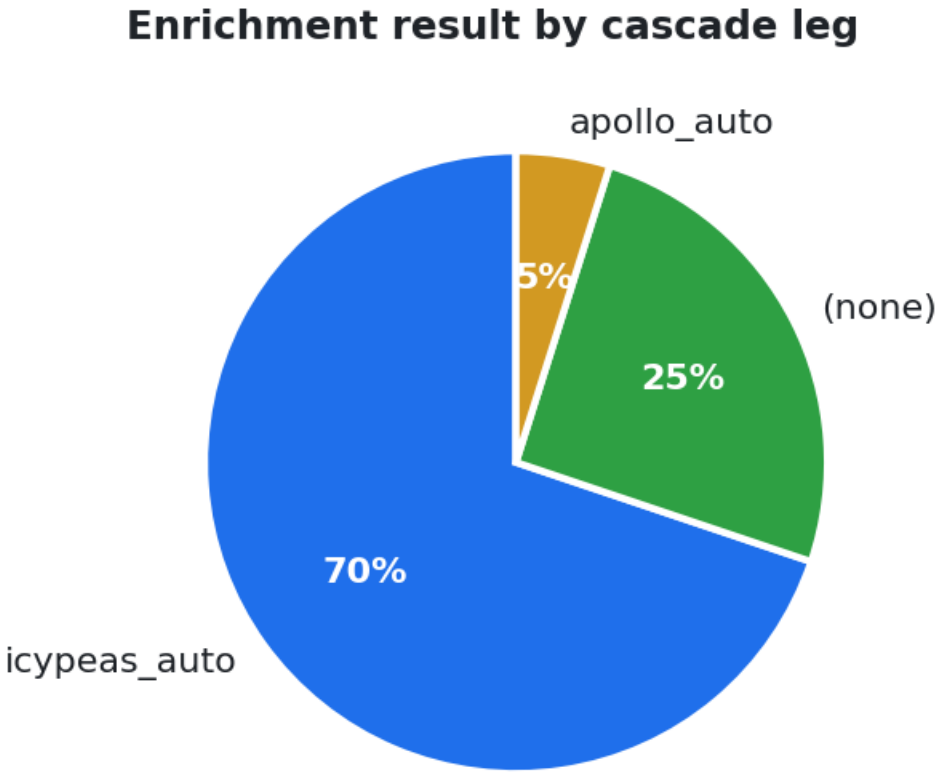
5 · Jobs and news page coverage

After a company enters the directory, the pipeline resolves its website and then probes common NL/EN sub-paths for a careers page and a news/blog page. Website resolution succeeds for **783 of 1,190 companies (66%)**; the remainder are either enrichment-terminal (no data from any provider) or brand-only entries without a domain.



Among the 783 companies with a discovered website, **430 (55%) have a jobs page**, **516 (66%) have a news / blog page**, and **357 companies have both**. Companies that don't surface a careers or news page are typically small distributors and OEM resellers who publish neither.

Enrichment cascade breakdown

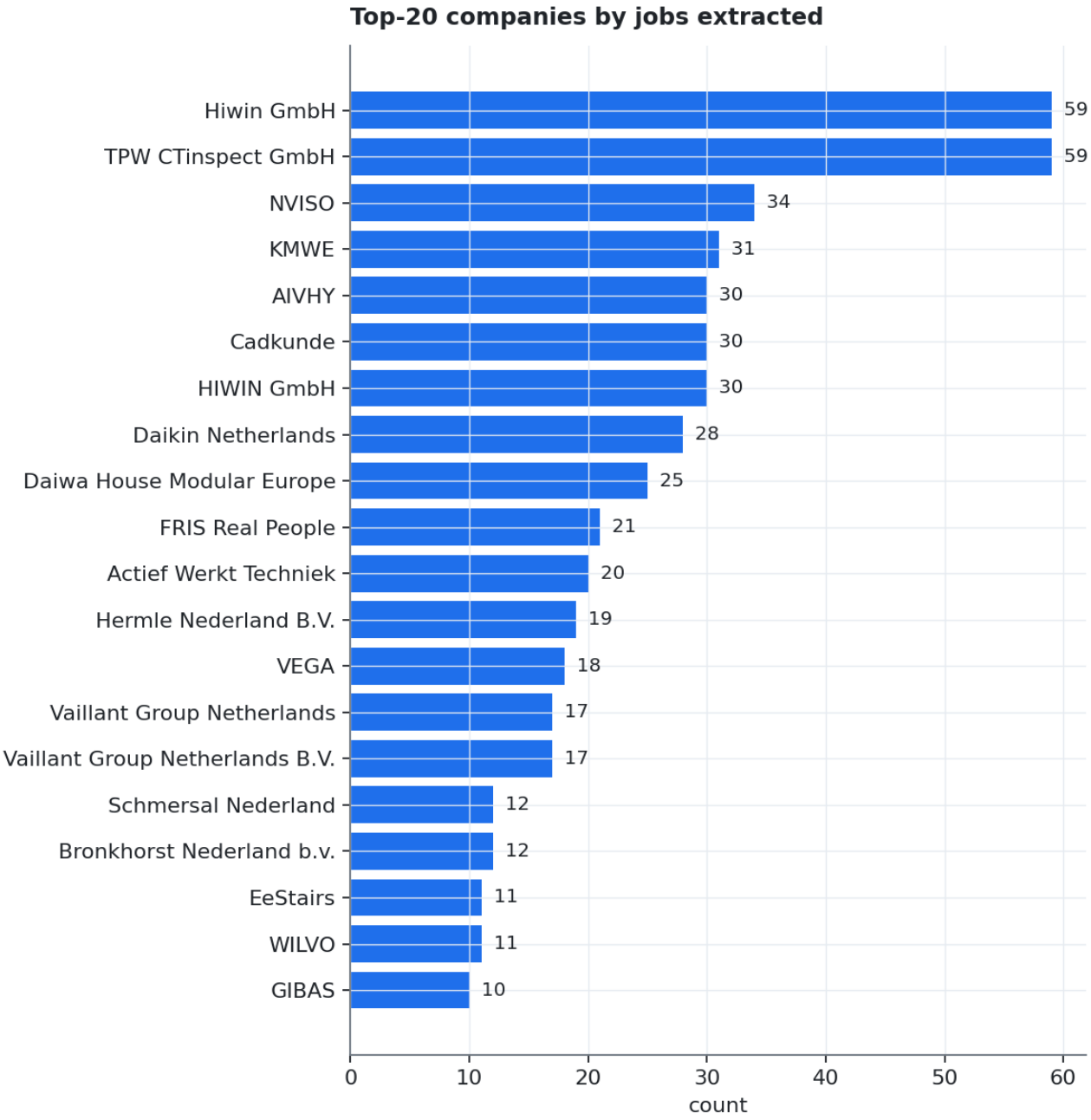


IcyPeas handles the overwhelming majority (831 companies), with Apollo picking up another 58 that IcyPeas couldn't hit. 301 companies are terminal (no data found by any of the four legs). FullEnrich and AI Ark are temporarily disabled behind feature flags and did not contribute to this snapshot.

6 · Jobs analysis

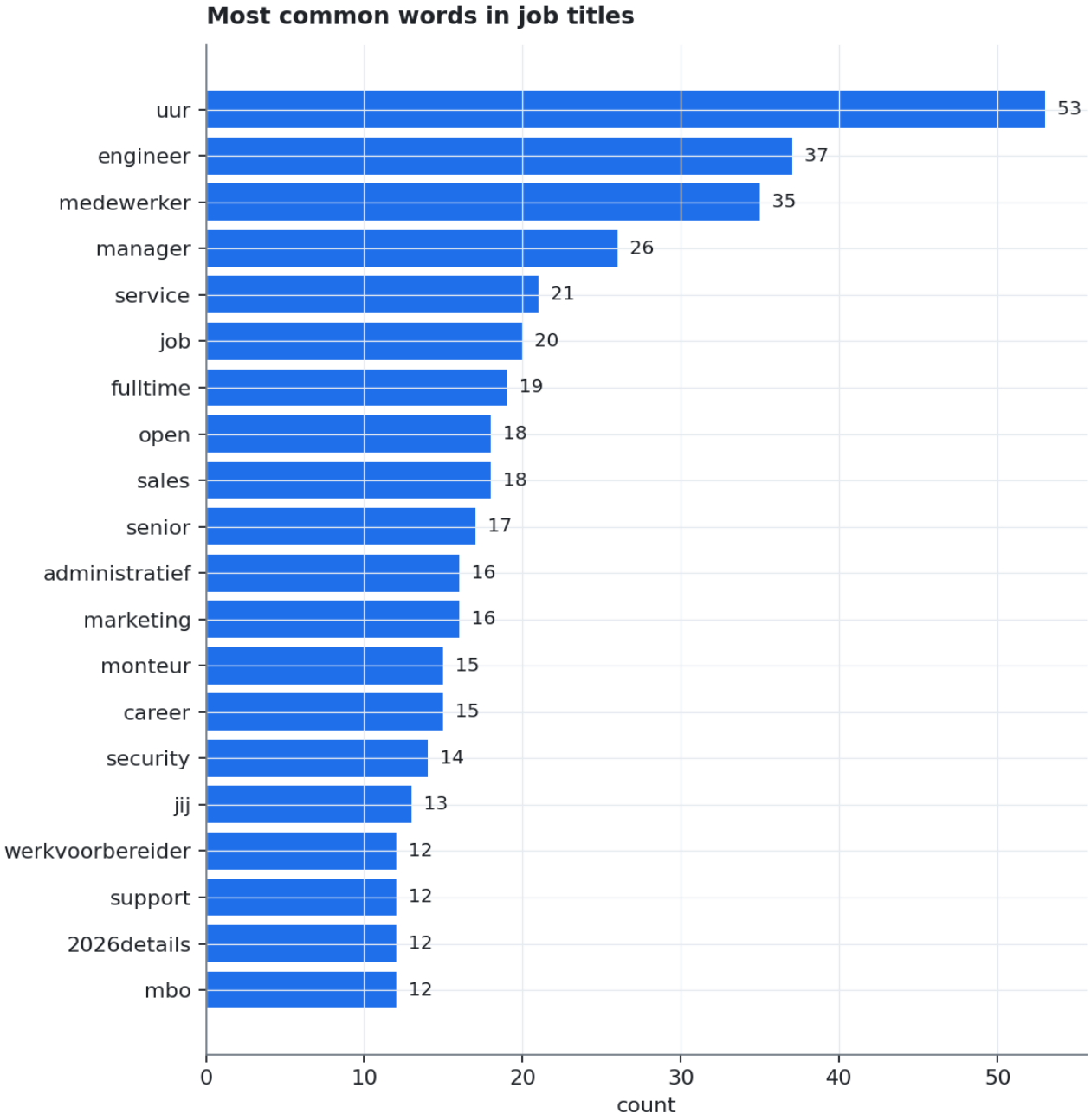
JOB LISTINGS	AVG / SOURCED COMPANY	TOP-5 CONCENTRATION	TOP-20 CONCENTRATION
930 across 430 sourced companies	**2.2** median lower (long tail)	**23%** 213 of total	**53%** 494 of total

6.1 · Top-20 companies by jobs found

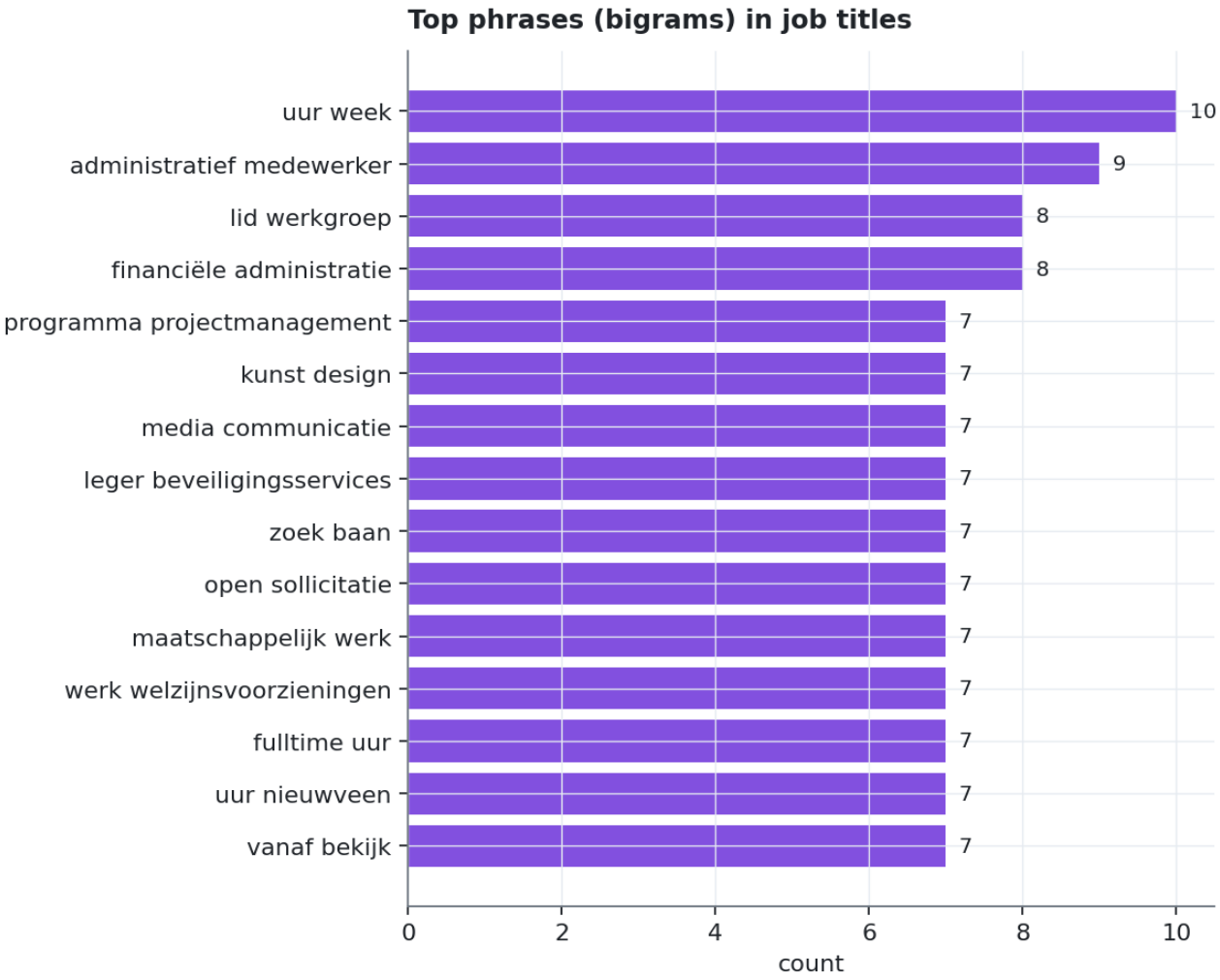


6.2 · Job-title word frequency

Word frequencies after stripping stop-words, month names and navigation-link residue (see caveat below). Job listings are technical: **engineer**, **manager**, **medewerker** (Dutch: *employee*) and **service** dominate.



6.3 · Top phrases (bigrams) in job titles

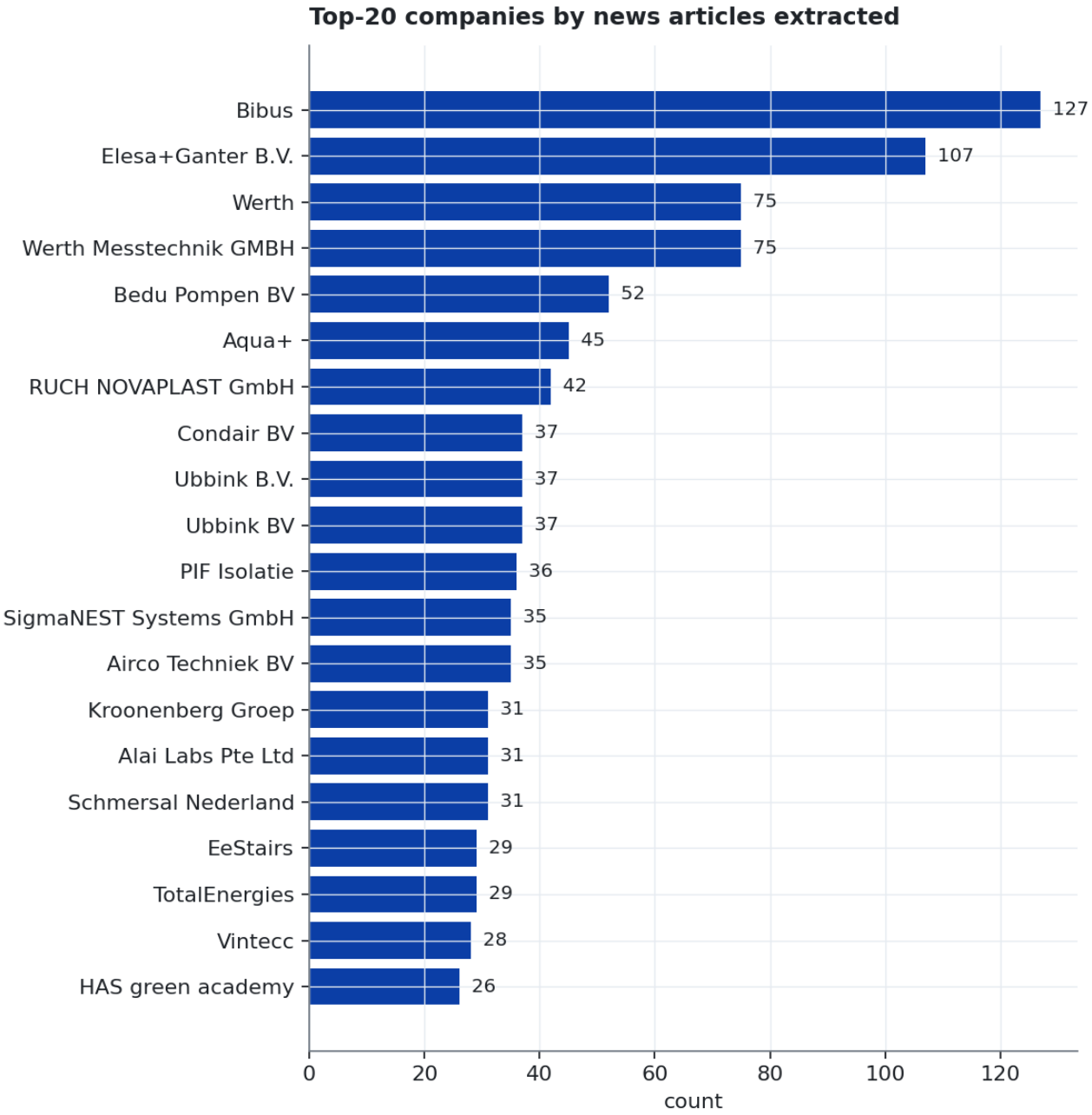


Caveats. The `jobs.department` and `jobs.location` columns are all `NULL` in the current snapshot — the listing extractor keeps only the title + URL. Categorisation is therefore derived from the title text only. Populating those columns from careers-page metadata would be a small extension. Additionally, 286 occurrences of navigation-link residue (*bekijk vacature*, *lees meer*, *learn more*, *read more*) were filtered out of the counts above — the raw title extractor also captures button/link text from surrounding page chrome.

7 · News / articles analysis

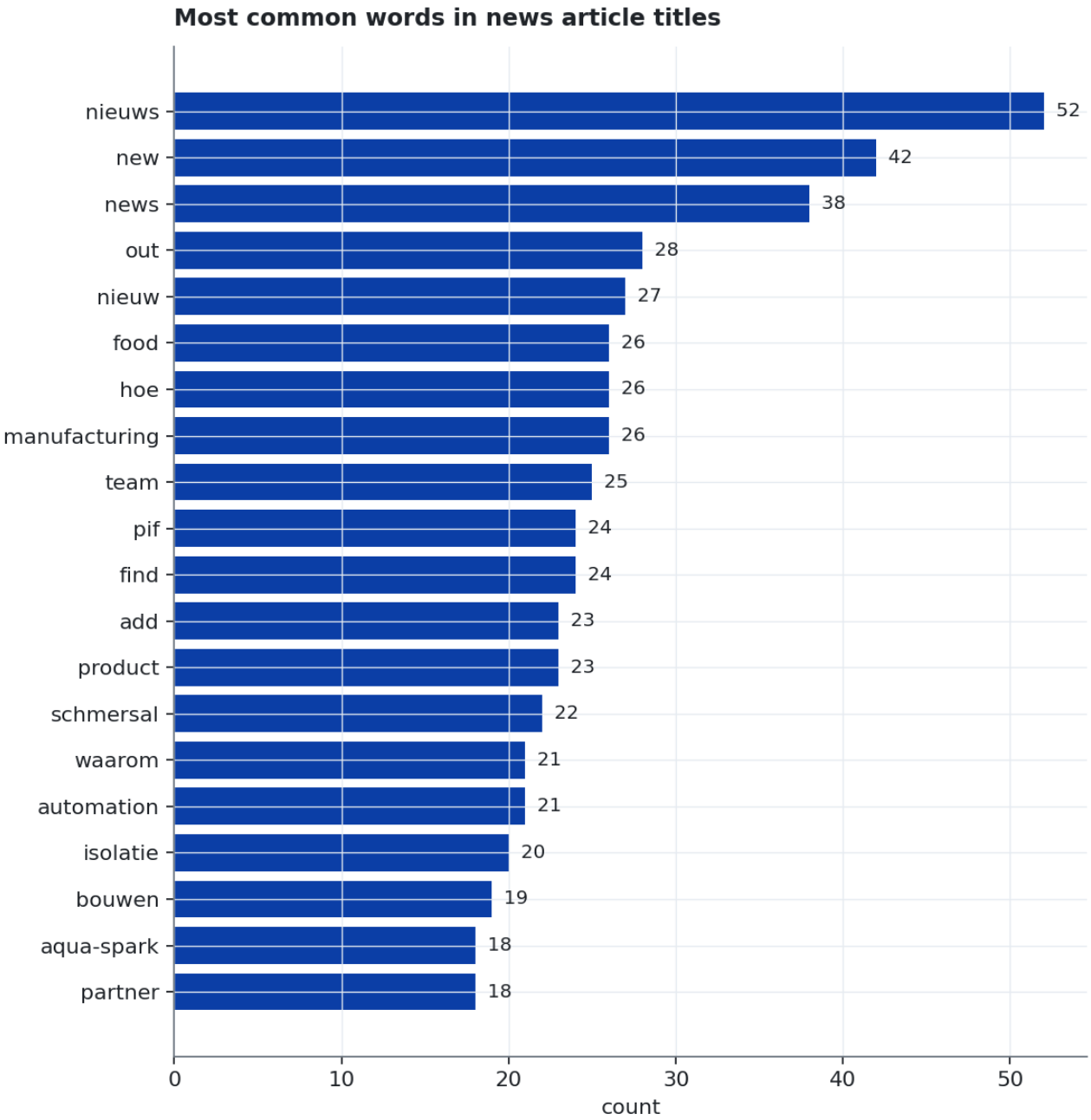
ARTICLES	AVG / SOURCED COMPANY	TOP-5 CONCENTRATION	TOP-20 CONCENTRATION
2,328 across 516 sourced companies	**4.5** median lower (long tail)	**19%** 436 of total	**41%** 945 of total

7.1 · Top-20 companies by news articles

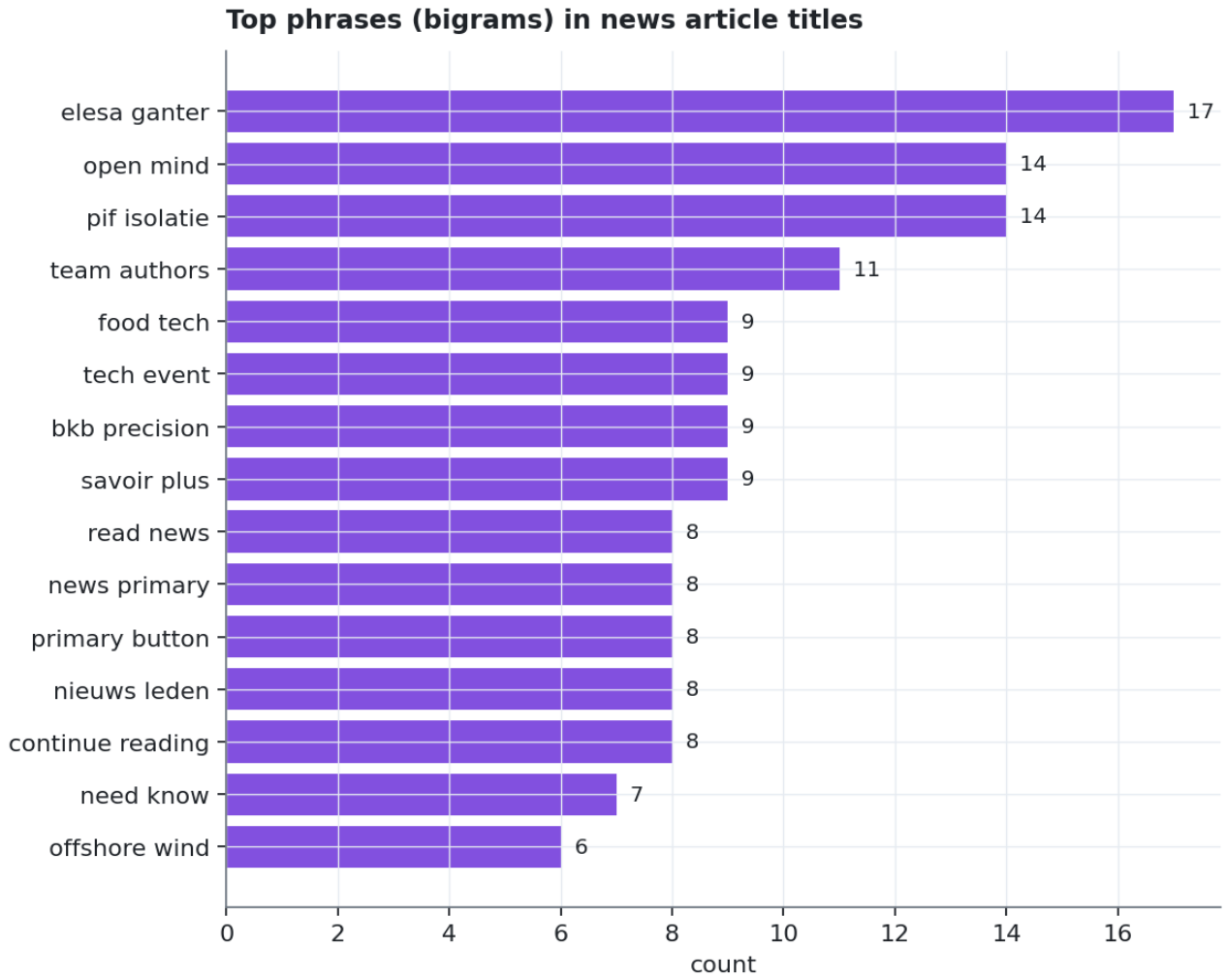


7.2 · Article-title word frequency

After filtering navigation residue and month names, the top words split into three groups: industry nouns (**nieuws**, **nieuwe**, **news**), multilingual chrome (**meer**, **lees** — Dutch *more* / *read*; **zum**, **blogbeitrag** — German blog-post header), and company/product names.



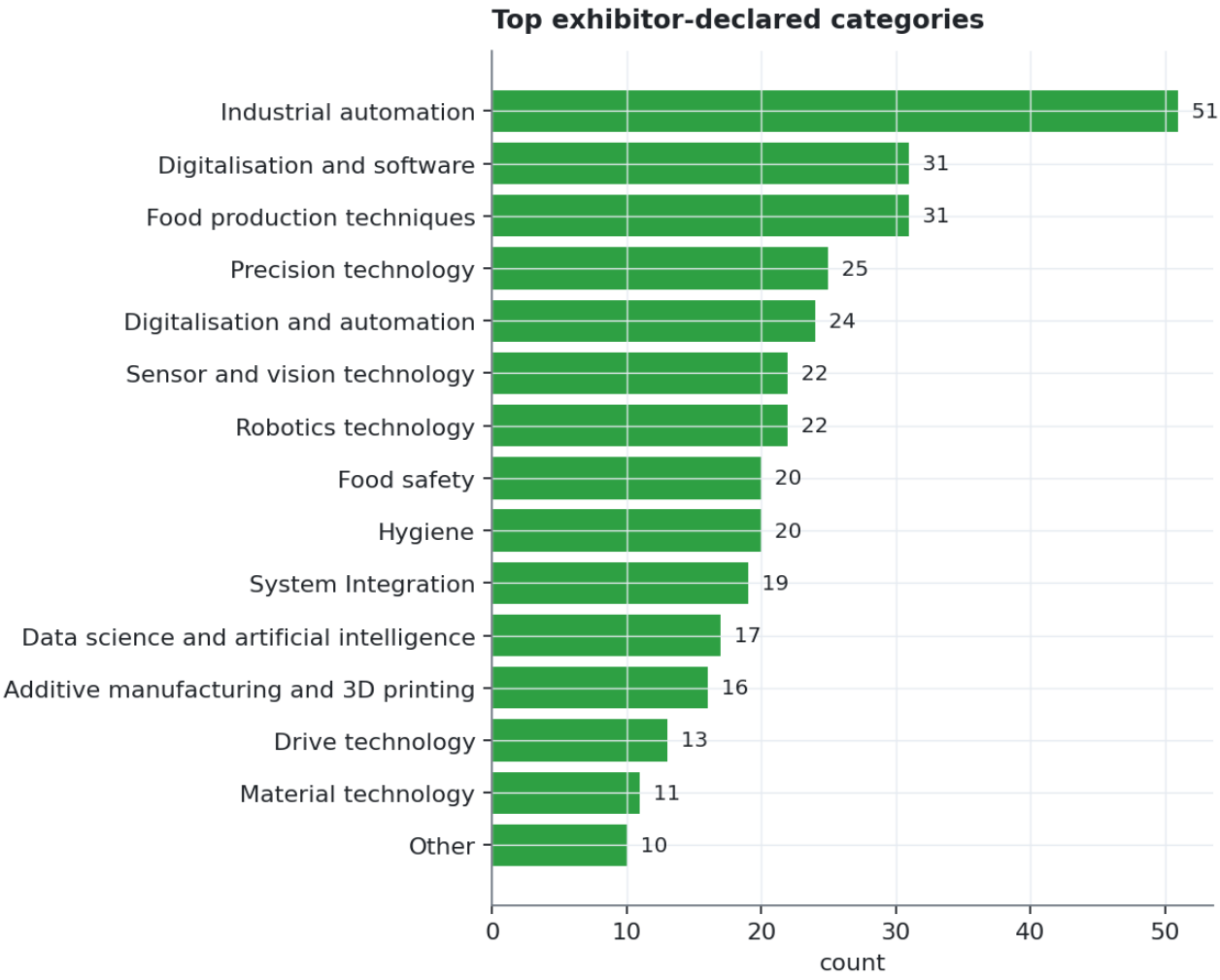
7.3 · Top phrases (bigrams) in article titles



Caveat. 676 single-word and 190 bigram occurrences of navigation-link residue were filtered out before ranking (*meer weten*, *lees meer*, *zum blogbeitrag*, *find out more*, *add arrow_forward*). These are "read more" buttons and Material Icons text that the anchor-based listing extractor picks up when the site puts them adjacent to article links. Fixing this would need a per-site excluder in `listing_extractor.py`.

8 · Exhibitor-declared categories

Trade-show pages that expose category tags on the exhibitor cards get them carried through into `category_1..3` on `companies`. Total distinct categories observed: **24**. The top 15 dominate the list.



Full category list (with counts)

Category	Companies
Industrial automation	51
Digitalisation and software	31
Food production techniques	31
Precision technology	25
Digitalisation and automation	24
Sensor and vision technology	22
Robotics technology	22
Food safety	20
Hygiene	20
System Integration	19
Data science and artificial intelligence	17

Category	Companies
Additive manufacturing and 3D printing	16
Drive technology	13
Material technology	11
Other	10
Robotics	9
Joining techniques	7
Metrology and analysis techniques	7
Maintenance technology	7
Sustainability	7
Optical and photonics technology	6
Electrical engineering	5
Energy management	4
Mechatronics technology	2

9 · Appendix — full ranking tables

9.1 · Top-25 companies by jobs found

#	Company	Jobs
1	Hiwin GmbH	59
2	TPW CTinspect GmbH	59
3	NVISO	34
4	KMWE	31
5	AIVHY	30
6	Cadkunde	30
7	HIWIN GmbH	30
8	Daikin Netherlands	28
9	Daiwa House Modular Europe	25
10	FRIS Real People	21
11	Actief Werkt Techniek	20
12	Hermle Nederland B.V.	19
13	VEGA	18
14	Vaillant Group Netherlands	17
15	Vaillant Group Netherlands B.V.	17
16	Schmersal Nederland	12
17	Bronkhorst Nederland b.v.	12
18	EeStairs	11
19	WILVO	11
20	GIBAS	10
21	Ubbink BV	10
22	Werth	10
23	HAS green academy	10
24	Summa Techniek	10
25	Veeneman B.V.	10

9.2 · Top-25 companies by news articles

#	Company	Articles
1	Bibus	127
2	Elesa+Ganter B.V.	107
3	Werth	75
4	Werth Messtechnik GMBH	75

#	Company	Articles
5	Bedu Pompen BV	52
6	Aqua+	45
7	RUCH NOVAPLAST GmbH	42
8	Condair BV	37
9	Ubbink B.V.	37
10	Ubbink BV	37
11	PIF Isolatie	36
12	SigmaNEST Systems GmbH	35
13	Airco Techniek BV	35
14	Kroonenberg Groep	31
15	Alai Labs Pte Ltd	31
16	Schmersal Nederland	31
17	EeStairs	29
18	TotalEnergies	29
19	Vintecc	28
20	HAS green academy	26
21	Cimatron	26
22	Allsetra BV	24
23	CLEVR	23
24	Ecochain	22
25	OPEN MIND Technologies Benelux b.v.	22

9.3 · Job-title vocabulary (top 30)

Word	Count
uur	53
engineer	37
medewerker	35
manager	26
service	21
job	20
fulltime	19
open	18
sales	18
senior	17
administratief	16

Word	Count
marketing	16
monteur	15
career	15
security	14
jij	13
werkvoorbereider	12
support	12
2026details	12
mbo	12
project	11
israel	11
wilvo	11
buitendienst	11
vacancies	10
germany	10
operations	10
employee	10
week	10
sollicitatie	9

9.4 · Article-title vocabulary (top 30)

Word	Count
nieuws	52
new	42
news	38
out	28
nieuw	27
food	26
hoe	26
manufacturing	26
team	25
pif	24
find	24
add	23
product	23
schmersal	22

Word	Count
waarom	21
automation	21
isolatie	20
bouwen	19
aqua-spark	18
partner	18
open	18
tot	18
award	18
min	18
elesa	18
ubbink	18
smart	17
samenwerking	17
weiterlesen	17
condair	17